Kolmogorov–Arnold Network for Remote Sensing Image Semantic Segmentation

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***Abstract*—Semantic segmentation is vital for remote sensing applications, where accurate feature extraction and representation are essential. Existing encoder–decoder architectures often fail to fully utilize high-dimensional features and recover fine details during decoding. To address this problem, we propose DeepKANSeg, a novel network based on the Kolmogorov–Arnold network (KAN). KAN's ability to decompose complex functions into univariate transformations enables flexible modeling of intricate data patterns. Our approach introduces two key innovations: a KAN-based deep feature refinement (DFR) module composed of DeepKAN to capture complex spatial and semantic relationships from high-dimensional features, and a global–local KAN (GLKAN) module replacing the traditional multilayer perceptron (MLP) layers with KAN-based linear layers to enhance fine-grained decoding. To evaluate the effectiveness of the proposed method, extensive experiments are conducted on two well-known fine-resolution remote sensing benchmark datasets, namely ISPRS Vaihingen and ISPRS Potsdam. The results demonstrate that the KAN-enhanced segmentation model achieves superior performance in terms of accuracy compared to state-of-the-art methods. Moreover, the univariate decomposition improves interpretability, making it suitable for explainable learning in remote sensing. The source code for this work will be accessible at https://github.com/sstary/SSRS**

***Index Terms*—Kolmogorov–Arnold networks (KANs), remote sensing, semantic segmentation.**

I. Introduction

SEMANTIC segmentation of remote sensing images is a fundamental task in Earth observation [1], forming the foundation for various downstream applications, such as environmental monitoring [2], [3], [4], disaster management [5], [6], [7], and urban planning [8], [9], [10]. To achieve efficient and accurate semantic segmentation, researchers have increasingly adopted deep learning methods for the automatic interpretation of remote sensing data. Early studies predominantly relied on convolutional neural networks (CNNs), renowned for their ability to capture local features and spatial hierarchies effectively. The transformer-based approach then gained traction due to its attention mechanism and ability to capture long-range contextual dependencies. More recently, the Mamba model has attracted attention for its low computational complexity in semantic segmentation tasks. Overall, these deep learning methods have greatly improved segmentation performance, enabling more robust and accurate analysis of remote sensing images.

More specifically, CNN-based methods were first proposed to address the challenges of high-resolution remote sensing image analysis. In particular, Long et al. [11] introduced the fully convolutional network (FCN), enabling CNNs to perform dense predictions and achieve pixel-level semantic segmentation. Later, the UNet model was proposed in [12], featuring a symmetric encoder–decoder architecture and skip connections to mitigate feature loss during forward propagation, thereby improving segmentation accuracy. More recently, a significant number of CNN-based approaches have been developed to tackle remote sensing semantic segmentation problems. More recently, a significant number of CNN-based methods have been proposed to tackle remote sensing semantic segmentation problems [13], [14], [15]. At the same time, the research community has increasingly recognized that the success of CNNs in dense prediction tasks is largely due to their convolutional operations, which focus on local features [16], [17], [18]. However, a key challenge remains in effectively capturing distant contextual dependencies, limiting their ability to model long-range interactions and broader contextual information.

In contrast, transformer-based models have been introduced into the field of computer vision [19], [20], [21], leveraging multihead self-attention (MSA) to capture global contextual information and model long-range dependencies. This attention mechanism effectively addresses the limitations of CNNs in modeling global dependencies. Motivated by these advancements, several studies have explored the application of transformer-based models to remote sensing image semantic segmentation [22], [23], [24], [25], achieving notable successes. While transformers are known for their powerful ability to capture global context, their computational process demands substantial memory and computational resources. As a result, researchers are often forced to make tradeoffs between the computational cost of the model and its feature extraction

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capability, which limits the practical application of transformer models.

More recently, the Mamba model [26], [27], which is developed upon the state space model (SSM), has attracted much attention because of its ability to model long-distance relationships with linear complexity [26]. In the field of remote sensing, RSMamba [28], RS-Mamba [29], and RS^{3}Mamba [30] introduced Mamba into remote sensing image semantic segmentation for the first time. Later, PyramidMamba [31] and PPMamba [32] integrated the pyramid pooling mechanism and selective scan module, proving that this hybrid structure effectively captures both local and global features of high-resolution remote sensing images. As a novel model architecture, Mamba still holds significant potential that remains to be further explored.

However, most existing deep learning methods discussed above rely on multilayer perceptron (MLP)-based computations, which struggle with handling complex high-dimensional features. Specifically, MLPs rely on global linear transformations and fixed activation functions for feature processing, which may not effectively capture intricate local and nonlinear relationships in high-dimensional data. Recently, the Kolmogorov–Arnold network (KAN) has emerged as a promising alternative to MLPs and has shown potential in several segmentation tasks due to its flexible nonlinear modeling capability [33], [34]. However, in complex high-resolution remote sensing tasks, the processing and restoration of high-dimensional semantic features depend heavily on fine-grained operations. Due to the complex semantic composition and fine spatial structures of remote sensing imagery, task-specific KAN-based designs for high-resolution remote sensing semantic segmentation remain insufficiently explored. In particular, how to effectively integrate KAN into both high-level semantic refinement and fine-grained spatial recovery remains an open problem.

To address the aforementioned challenges, we propose to leverage KAN to decompose high-dimensional functions into a combination of univariate functions, offering a flexible mechanism for modeling complex and localized feature relationships, which is particularly beneficial for remote sensing scenes with intricate spatial structures and high-dimensional semantic interactions [35]. Recognizing the significance of pretrained models for feature extraction, we utilize KAN in two key stages of the encoder–decoder architecture: high-level semantic refinement and fine-grained spatial recovery. Specifically, we propose the DeepKAN module to further refine rich semantic information generated by the encoder. Through nonlinear operations along the channel dimension, the semantic information embedded in abstract high-dimensional features is effectively utilized. In addition, we present a global–local KAN (GLKAN) module composed of KAN-based linear layers to replace the MLP operations in the decoder for enhanced fine-grained decoding. This facilitates the stepwise restoration of spatial and semantic information, ultimately enabling precise pixel-level classification. The main contributions of this work are summarized as follows.

1) We first propose the DeepKAN architecture that offers significantly enhanced interpretability and refined abstract semantic features by extending the KAN's ability to decompose high-dimensional data into learnable univariate transformations in remote sensing.
2) Leveraging the capabilities of the DeepKAN architecture, a novel deep feature refinement (DFR) module is devised. This module is designed to fully harness the intricate, high-dimensional semantic space inherent in remote sensing data. By employing nonlinear transformations along channel dimensions, it effectively captures complex spatial and semantic relationships, enabling precise feature enhancement.
3) A decoder based on GLKAN is introduced by exploiting a unique global–local combined attention mechanism. This design progressively restores detailed ground category information through concise nonlinear operations, ensuring accurate reconstruction of semantic features while maintaining computational efficiency.
4) By integrating the DeepKAN-based DFR module and the GLKAN-based decoder, a new semantic segmentation network called DeepKANSeg is developed. This architecture combines DFR and precise decoding, offering a robust and scalable solution for high-resolution remote sensing image segmentation tasks.

The adoption of KAN in this work is expected to pave the way for a novel and highly promising research direction, offering substantial potential for further advancements in this field. The structure of this article is as follows. Section II first reviews related work on semantic segmentation and KAN in remote sensing. Section III then presents the details of the proposed DeepKANSeg, followed by Section IV, which offers a comprehensive analysis of the experimental results. Finally, Section V concludes this article.

II. Related Works

A. Remote Sensing Semantic Segmentation

Semantic segmentation has achieved remarkable advancements in the field of remote sensing. CNNs have served as foundational tools in remote sensing image analysis, owing to their robust feature extraction capabilities [36], [37]. Models such as PPResNet [38], DeepResUnet [39], and ResUNet-a [40] have demonstrated strong performance in semantic segmentation tasks for remote sensing images. These approaches systematically examined the effects of various convolutional kernels and network depths on task performance. In particular, ResUNet-a [40] introduced a conditioned relationship between different tasks, enhancing the model's convergence efficiency. Despite their success, CNNs are inherently limited by their local receptive field and inability to effectively capture global contextual information. This limitation incurs significant information loss when processing complex remote sensing scenarios, particularly in large-scale or high-resolution images.

To address the aforementioned challenges, the Transformer architecture has been introduced in recent years. Its ability to effectively capture global information makes it particularly well-suited for processing large-scale images in remote sensing semantic segmentation [23], [24], [41], [42], [43], [44], [45],

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[46], [47]. For example, UNetFormer [23] incorporated an efficient global–local attention mechanism to enhance the decoder's ability to interpret global and local information. Similarly, FTransUNet [45] addressed the multimodal fusion challenge by combining a fusion vision Transformer with CNN blocks to extract local features across different modalities. Despite these advancements, the high computational complexity of transformer-based models limits their broader adoption in high-resolution remote sensing imagery, particularly in computational resource-constrained environments.

Mamba is an emerging lightweight deep learning framework that leverages the SSM to balance computational efficiency and accuracy [48], [49], [50], making it highly attractive for remote sensing semantic segmentation tasks [28], [29], [30], [51], [52]. For instance, RS-Mamba [29] introduced an omnidirectional selective scan technique to capture large spatial features from multiple directions, expanding the scanning capabilities of Mamba for a more comprehensive and effective learning process. Meanwhile, RS^{3}Mamba [30] used auxiliary CNN layers to extract local features while using Mamba to model long-range dependencies. These approaches represent two distinct strategies for introducing Mamba into remote sensing applications.

In general, the aforementioned structures often utilize MLP as a fundamental component. Recently, the KAN has been proposed as a promising alternative to MLP, offering the potential to introduce new learning capabilities [33], [34], [53]. However, the application of KAN to remote sensing semantic segmentation for high-resolution imagery remains unexplored, presenting a compelling opportunity for future research. Its potential to improve semantic segmentation in remote sensing data necessitates thorough validation through detailed experiments and further studies.

B. Kolmogorov–Arnold Network

KAN [33] is a novel neural network architecture designed to replace the traditional MLP structure by utilizing a learnable function in place of a fixed activation function, leveraging the Kolmogorov–Arnold theorem. This theorem states that any multivariable continuous function can be represented as a combination of a finite number of single-variable continuous functions. KAN operationalizes this by decomposing complex high-dimensional functions into a set of simple 1-D functions, typically parameterized as spline functions. This approach offers exceptional flexibility and enables the modeling of complex functions, while also enhancing the model's interpretability.

KAN has been initially applied in the fields of computer vision [54] and medical imaging [55], [56], demonstrating its versatility and effectiveness. In computer vision, Convolutional KAN [57] integrated KAN into convolutional layers by implementing spline functions as kernels. This approach outperforms traditional convolutional layers, resulting in a more adaptable and learnable network. In medical imaging, U-KAN [55] incorporated KAN layers into the U-Net framework. By introducing nonlinear learnable activation functions, it achieved improved accuracy and interpretability. In addition, TransUKAN [58] leveraged KAN to capture nonlinear relationships with minimal additional parameters, addressing the transformer's limitations in extracting local information while reducing memory usage and computational overhead.

In the field of remote sensing, the application of KAN is still in its early stages. Preliminary studies highlight the potential of KAN in remote sensing optical image processing [59], [60], [61], [62], [63], [64] and hyperspectral image classification [65], [66], [67], [68]. In particular, Cheon [59] replaced traditional MLP layers with KAN for remote sensing image classification and explored its combination with various pretrained CNN and ViT models. The results reveal that KAN enhances classification accuracy while reducing computational complexity. Furthermore, Liu et al. [60] marked the first use of pure KAN in remote sensing multimodal change detection tasks, with experimental results demonstrating its effectiveness and superiority. Wav-KAN [67] introduced wavelet functions as learnable activation mechanisms, facilitating the nonlinear mapping of input spectral signatures. These studies demonstrate the potential of KAN in remote sensing image analysis, but most of them focus on scene classification, change detection, or hyperspectral image analysis. In contrast, high-resolution remote sensing semantic segmentation requires both high-level semantic refinement and fine-grained spatial recovery for dense pixel-level prediction. Therefore, task-specific KAN-based designs for remote sensing semantic segmentation remain insufficiently investigated. Different from existing KAN-based segmentation methods mainly designed for medical or industrial inspection images, this work explores how KAN can be integrated into deep semantic refinement and global–local decoding for high-resolution remote sensing imagery.

III. Methodology

A. Preliminary: KAN

This work aims to develop a general framework for integrating KAN into remote sensing semantic segmentation, building on its demonstrated efficiency and interpretability as highlighted in [33]. KAN is inspired by the Kolmogorov–Arnold representation theorem, which provides a theoretical foundation for expressing any multivariable continuous function as a combination of single-variable continuous functions [69]. This theorem is expressed as

$$f(\boldsymbol{x}) = \sum_{q} \phi_q \left(\sum_{p} \psi_{pq} \left(x_p \right) \right) \tag{1}$$

where $rp = 1, 2, \ldots, n_{\text{in}}$ and $q = 1, 2, \ldots, n_{\text{out}}$ with n_{in} and n_{out} denoting the dimension of input and output features, respectively. In addition, $\boldsymbol{x}$ represents the input vector with x_p being the pth element of $\boldsymbol{x}$. Finally, ϕ_q and ψ_{pq} are continuous single-variable functions.

In contrast to MLPs shown in Fig. 1(a), which rely on fully connected layers with fixed activation functions, KAN, shown in Fig. 1(b), distinguishes itself by using learnable and parameterized activation functions. Specifically, the learnable

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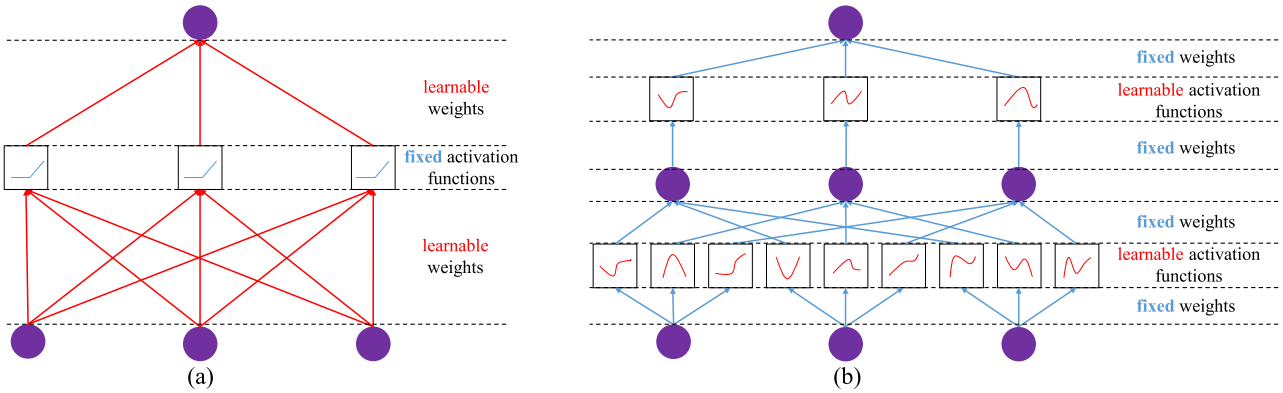


Fig. 1. Structure of (a) MLP and (b) two-layer KAN follows (1). The former learns by adjusting the learnable weights using fixed activation functions. In contrast, the latter learns through adjusting the learnable activation functions using fixed weights.

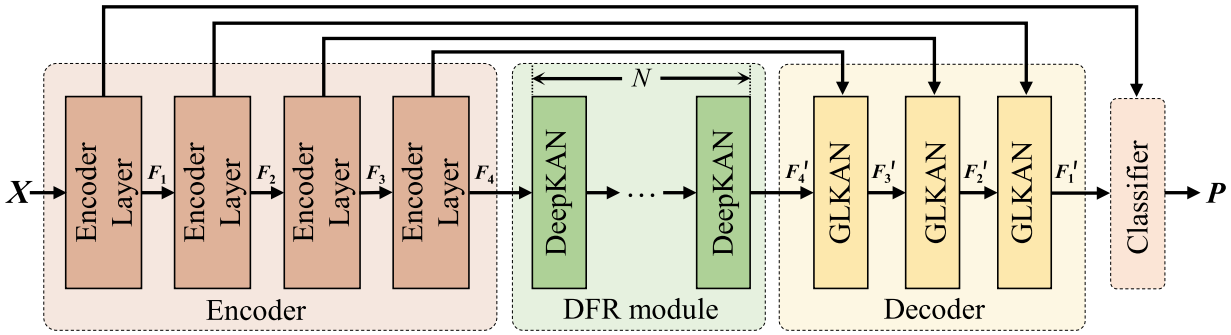


Fig. 2. Overview of the proposed DeepKANSeg, which is comprised of three parts: an encoder, a DFR module, a decoder, and a classifier. The combination of a pretrained encoder with the KAN-based DFR module and decoder further enhances the semantic segmentation performance.

univariate functions ψ_{pq} and ϕ_q, are parameterized by B-spline basis functions as

$$\psi_{pq}(x_p) = w^b_{pq} b(x_p) + \sum_{\ell=1}^{G+r} w^s_{pq,\ell} B^{(r)}_\ell(x_p)$$
$$\phi_q(u) = \bar{w}^b_q b(u) + \sum_{\ell=1}^{G+r} \bar{w}^s_{q,\ell} B^{(r)}_\ell(u) \tag{2}$$

where u denotes the scalar input of $\phi_q(\cdot)$, $b(\cdot)$ denotes the base activation function, $B^{(r)}_\ell(\cdot)$ is the ℓth B-spline basis function of order r, and w^b_{pq}, $\bar{w}^b_q$, $w^s_{pq,\ell}$, and $\bar{w}^s_{q,\ell}$ are learnable coefficients. Here, G represents the grid size, i.e., the number of grid intervals used to discretize the input range, while r denotes the spline order. By parameterizing ϕ_q and ψ_{pq} as (2), KAN enables highly flexible modeling of complex relationships without using traditional linear weight matrices. In addition, the explicit univariate decomposition enhances interpretability, providing deeper insights into the model's internal decision-making processes.

Liu et al. [33] also presented the deep network architecture incorporating multiple KAN layers, each composed of a matrix of learnable 1-D functions, enabling flexible modeling of complex feature interactions. Mathematically, a network of K KAN layers can be modeled as

$$\mathrm{KAN}(\boldsymbol{x}) = (\boldsymbol{\Phi}_{K-1} \circ \boldsymbol{\Phi}_{K-2} \circ \cdots \circ \boldsymbol{\Phi}_1 \circ \boldsymbol{\Phi}_0)\boldsymbol{x} \tag{3}$$

where $\boldsymbol{\Phi}_k$ denotes a KAN layer for $k = 0, 1, \ldots, K-1$, and $\circ$ stands for the composition operation. This deep network architecture can be seamlessly integrated into existing semantic segmentation networks for remote sensing tasks, offering a powerful solution to enhance feature representation learning.

B. Proposed DeepKANSeg

In this work, we introduce KAN into general semantic segmentation methods for remote sensing by developing an innovative framework based on the classical encoder–decoder architecture. The proposed DeepKANSeg, illustrated in Fig. 2, consists of four primary components: an encoder, a DFR module, a decoder, and a classifier. As there are no pretrained KAN-based models available, which could significantly impact segmentation performance, we instead utilize well-established CNN or transformer-based encoders. During both the training and test stages, the encoder extracts multiscale features from the input. These features are subsequently processed by the DFR module, which leverages stacked DeepKAN modules to effectively exploit high-dimensional abstract representations. The refined features are then passed through a decoder comprising GLKAN modules, which iteratively reconstruct semantic information with corresponding skip connections from the encoder. Notably, the weighted sum for handling skip connections is omitted for the sake of clarity in the figures. Finally, a classifier generates the pixel-level segmentation map.

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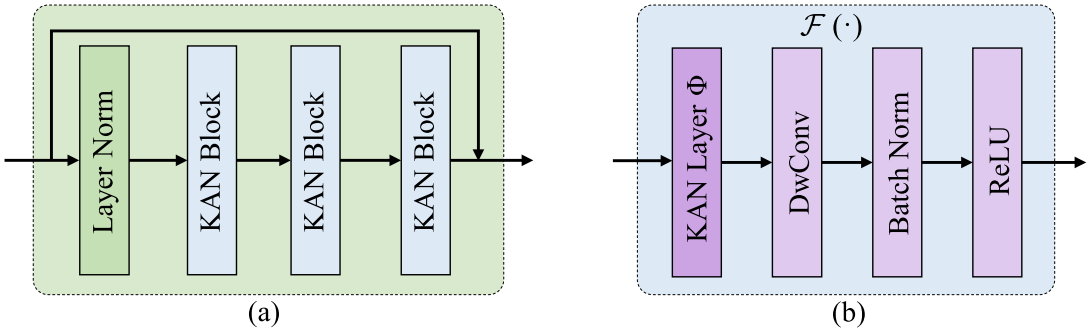


Fig. 3. (a) Structure of the proposed DeepKAN and (b) structure of the KAN block. The high-dimensional feature is refined by the stacked KAN layer, which is crucial for extracting complex remote sensing semantic information.

C. Encoder

We denote the input as $\boldsymbol{X} \in \mathbb{R}^{H\times W\times C}$, where H, W are the spatial dimensions, and C is the number of channels (e.g., $C = 3$ for RGB images). The encoder progressively extracts feature maps with different scales, denoted as $\boldsymbol{F}_i \in \mathbb{R}^{H/(2^{i+1})\times W/(2^{i+1})\times C_i}$, where C_i is the number of channels at stage $i = \{1, 2, 3, 4\}$. In this work, CNN-based ResNet [70], and transformer-based plain ViT [71], [72] are utilized to demonstrate the flexibility and robustness of the proposed DeepKANSeg across different backbone designs.

1) ResNet: The input is initially processed through a convolutional layer, followed by a max-pooling operation that reduces the spatial dimensions. The core network consists of four stages of residual blocks, where each stage doubles the channel dimensions and reduces the spatial resolution using convolutions. Each residual block contains several convolutional layers with skip connections, and they finally produce multiscale feature maps $\boldsymbol{F}_i$.

2) Plain ViT: The input is divided into nonoverlapping patches of size $P \times P$. Each patch is flattened into a vector, forming a sequence of patch embeddings $\boldsymbol{Z}_0 \in \mathbb{R}^{L\times D}$, where $L = (H \times W)/(P^2)$ represents the number of patches, and D is the embedding dimension. Positional encodings are added to retain spatial information. The core network consists of a series of transformer layers, each comprising an MSA module and a feed-forward network (FFN), both equipped with layer normalization (LN) and residual connections. The output of the final transformer layer is reshaped to $\boldsymbol{F}_p \in \mathbb{R}^{H/P\times W/P\times D}$ and then used to generate multiscale feature maps through pyramid modules, which include parallel convolutions or deconvolutions [72]. This simple approach effectively generates multiscale features $\boldsymbol{F}_i$.

D. DFR Module Empowered by DeepKAN

The high-dimensional feature $\boldsymbol{F}_4$ extracted by the encoder is processed through the proposed DFR module composed of stacked DeepKAN modules. It can effectively learn abstract semantic features through fine-grained learnable activation functions. Initially, $\boldsymbol{F}_4$ is flattened into serialized data $\boldsymbol{Z}_{\text{in}}$, and then passed through N DeepKAN modules. As shown in Fig. 3(a), each DeepKAN module consists of an LN and three KAN blocks. Considering the richness of high-dimensional channel information, a depth-wise convolution operation along the channel dimension is employed to assist the learning process of KAN. As depicted in Fig. 3(b), a KAN block includes a KAN layer, an efficient depth-wise convolutional layer (DwConv), a batch normalization layer (BN), and a ReLU activation function [55]. The process of KAN block, denoted as $\mathcal{F}(\cdot)$, can be expressed as

$$\mathcal{F}(\boldsymbol{Z}_{in}) = \sigma(\text{BN}(\text{DwConv}(\boldsymbol{\Phi}\boldsymbol{Z}_{in}))) \tag{4}$$

where $\sigma(\cdot)$ represents the ReLU activation. Furthermore, $\boldsymbol{\Phi}$ denotes the KAN layer whose output serves as the input for the subsequent KAN block. After three consecutive KAN blocks, a complete DeepKAN process is completed.

After processing through N DeepKAN modules, the final $\boldsymbol{Z}_{out}$ is reshaped back to the original input size denoted as $\boldsymbol{F}'_4$. At this stage, the high-dimensional semantic information is further refined and enriched.

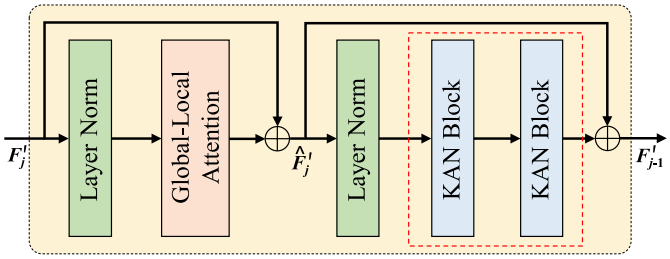


Fig. 4. Structure of the proposed GLKAN is illustrated. It resembles the classic transformer block, with the component highlighted in the red box modified into KAN-based blocks.

E. Decoder

The output from the DFR module is subsequently decoded using the proposed decoder, whose core component is GLKAN. As illustrated in Fig. 4, GLKAN adopts the structure of a typical transformer block, comprising an attention block and a fully connected block, each equipped with an LN. The attention block employs a classic global–local attention mechanism [23], which consists of two parallel branches: a global branch leveraging window-based self-attention to capture long-range dependencies [43] and a local branch using convolutional operations for local feature exploiting. The

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TABLE I
ILLUSTRATION OF FOUR DATASETS

	ISPRS Vaihingen	ISPRS Potsdam	LoveDA	SpaceNet3
Geographical Type	Small Town	Urban	Complex Scenes	Urban
Task Type	Semantic Seg.	Semantic Seg.	Semantic Seg.	Road Extraction
Spatial Resolution	9 cm	5 cm	30 cm	∼30 cm
Spectral Channel	NIRRG	RGB	RGB	RGB
Data Volume	16 patches	24 patches	2653	2780
Patch Size	2500 × 2000	6000 × 6000	1024	1300
Training/Test Sample Size	256	256	256	1024
Test Stride	32	128	64	-
Training-Test Ratio	3-1	3-1	5-2-3	4-1

outputs of these branches are fused via element-wise summation, enabling comprehensive extraction of both global and local contexts. GLKAN differs from the standard global–local transformer block by substituting the fully connected layer with KAN-based modules, allowing fine-grained channel-level operations to more effectively recover semantic features.

To preserve similarity with the original MLP structure in a classical transformer block [73], two KAN blocks are used in GLKAN. Mathematically, the process of GLKAN can be written as

$$\hat{\boldsymbol{F}}'_j = \mathrm{GLAttn}\left(\mathrm{LN}\left(\boldsymbol{F}'_j\right)\right) + \boldsymbol{F}'_j \tag{5}$$

$$\boldsymbol{F}'_{j-1} = \mathcal{F}\left(\mathcal{F}\left(\mathrm{LN}\left(\hat{\boldsymbol{F}}'_j\right)\right)\right) + \hat{\boldsymbol{F}}'_j \tag{6}$$

where GLAttn represents the global–local attention mechanism, $\mathcal{F}(\cdot)$ denotes the KAN block as defiend in (4), and the subscript $j = \{4, 3, 2\}$ is the number of channels at the decoding stage. The decoder progressively decodes the features, gradually recovering spatial information and revealing ground class details. Finally, $\boldsymbol{F}'_1$ is processed by a classifier to generate the segmentation map denoted by $\boldsymbol{P} \in \mathbb{R}^{H\times W\times C'}$, where C' is the number of classes.

In addition to reconstructing semantic features, the proposed GLKAN decoder also helps preserve fine spatial structures during the decoding process. By combining global–local attention with KAN-based nonlinear transformations, the decoder is able to capture both contextual dependencies and local spatial variations more effectively. The global branch models long-range relationships among objects, while the local convolutional branch focuses on spatially continuous structures. Meanwhile, replacing conventional MLP layers with KAN-based blocks enables more flexible nonlinear transformations along the channel dimension, which helps retain detailed structural information during feature reconstruction. As a result, the decoded feature maps contain richer spatial details, contributing to more accurate delineation of object boundaries in the final segmentation results.

IV. EXPERIMENTS AND DISCUSSION

A. Datasets

We evaluate the proposed method on four remote sensing datasets, namely ISPRS Vaihingen, ISPRS Potsdam, LoveDA, and SpaceNet3. The first three datasets are used for semantic segmentation, while SpaceNet3 is adopted for road extraction, which can be formulated as a binary segmentation task. All datasets cover complex scenes with diverse land-cover types. The details are summarized in Table I.

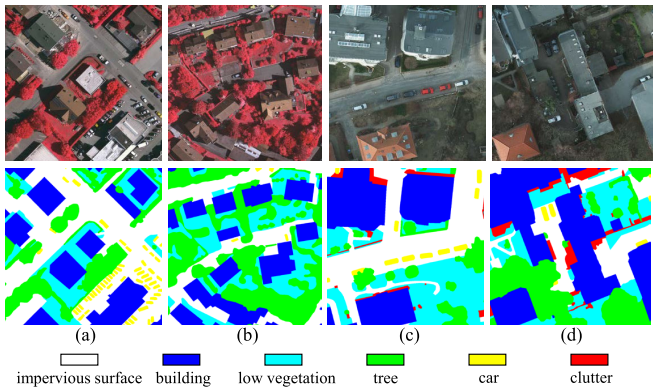


Fig. 5. Four visual samples of size 1024 × 1024 are presented: (a) and (b) show two different samples from the Vaihingen dataset, while (c) and (d) show two different samples from the Potsdam dataset. The first row contains the orthophotos, with Vaihingen represented in NIRRG channels and Potsdam in RGB channels, while the second row shows the corresponding ground truth. The diverse and complex urban features in these datasets make semantic segmentation particularly challenging. Experiments conducted on these two datasets provide a comprehensive evaluation of the model's performance.

To improve computational efficiency, we adopt a sliding window approach to extract samples. The training and test sample sizes for Vaihingen and Potsdam are both set to 256, with test strides of 32 and 128, respectively. For LoveDA, the sample size is set to 256 with a test stride of 64. For SpaceNet3, the sample size is set to 1024; random window sampling is adopted during training, while center cropping is used during testing. Fig. 5 illustrates representative samples from the datasets.

B. Evaluation Metrics

To evaluate the segmentation performance of the proposed DeepKANSeg, we utilize the mean F_1 (mF_1) and the mean Intersection over Union (mIoU) in our experiments. These widely adopted metrics enable a fair comparison of our method's performance against state-of-the-art methods. The mF_1 and mIoU values are calculated for the five foreground classes, while the class labeled as *Clutter* is treated as a background or sparse class, and hence, performance statistics for this class are not reported [24], [74].

For the SpaceNet3 dataset, which focuses on road extraction, we additionally report precision and recall to provide a

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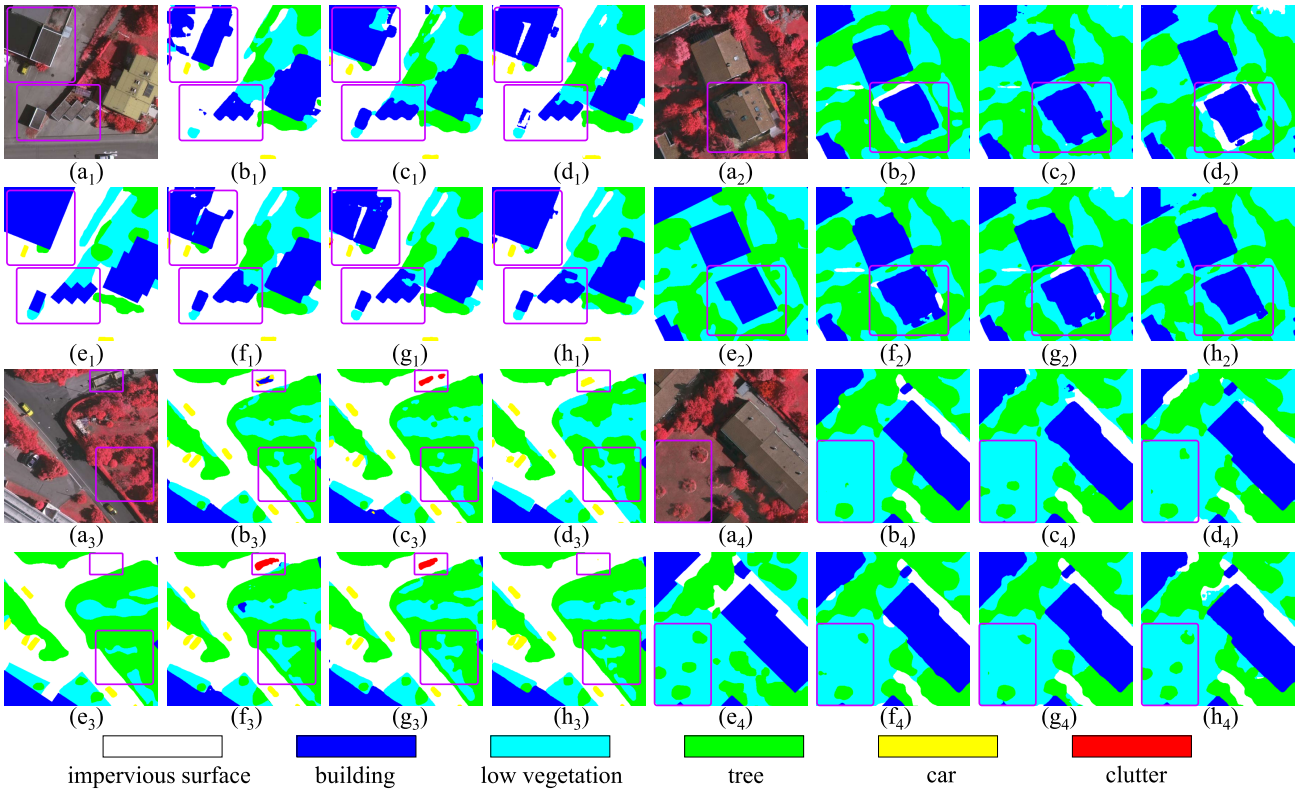


Fig. 6. Visualization results on the ISPRS Vaihingen dataset of size 512 × 512. (a) NIRRG images. (b) UNetformer. (c) FTUNetformer. (d) CMTFNet. (e) Ground truth. (f) SSNet. (g) RS^{3}Mamba. (h) Proposed DeepKANSeg. Some purple boxes are marked to highlight the differences.

more comprehensive evaluation of detection performance. The F_1 and IoU values for each class indexed by n are computed using the following formulas:

$$F_1 = 2 \times \frac{\text{precision}_n \cdot \text{recall}_n}{\text{precision}_n + \text{recall}_n} \tag{7}$$

$$\text{IoU} = \frac{TP_n}{TP_n + FP_n + FN_n} \tag{8}$$

where TP_n, FP_n, and FN_n are true positives, false positives, and false negatives for the nth class, respectively. The precision and recall are defined as

$$\text{precision}_n = \frac{TP_n}{TP_n + FP_n} \tag{9}$$

$$\text{recall}_n = \frac{TP_n}{TP_n + FN_n}. \tag{10}$$

After calculating the F_1 and IoU values for the foreground classes based on the definitions provided, their mean values, referred to as mF_1 and mIoU, can be obtained, respectively.

C. Implementation Details

The experiments were conducted using PyTorch on a single NVIDIA GeForce RTX 4090 GPU with 24 GB of RAM. All models were trained with the stochastic gradient descent algorithm, utilizing a learning rate of 0.01, a momentum of 0.9, and a weight decay coefficient of 0.0005. The total number of training epochs was set to 50, and the multistep learning rate schedule was applied with step values of $[25, 35, 45]$ and $\gamma = 0.1$. The batch size was set to 10. Simple data augmentations, such as random rotation and flipping, were applied after the sliding window collected the samples. The DFR module consists of four stacked DeepKAN modules, i.e., N = 4. As previously discussed, we performed experiments using two distinct pretrained backbones for DeepKANSeg, namely the CNN-based ResNet18 and the vision transformer-based ViT-L. For the latter, we utilized SAM's encoder [75] and fine-tuned it using Adapter [76], which can greatly reduce training costs. The differences in structure and scale between the two backbones facilitated a comprehensive evaluation of the proposed DeepKANSeg's effectiveness. Finally, all models in this study are optimized using the cross-entropy loss, without the introduction of any additional loss functions, to ensure the fairness of the comparative experiments.

D. Performance Comparison

We benchmarked the performance of the proposed DeepKANSeg against several representative state-of-the-art methods, namely ABCNet [22], TransUNet [77], UNetFormer [23], FTUNetFormer [23], CMTFNet [24], SSNet [46], and RS^{3}Mamba [30]. These methods cover a range of widely used architectures, including CNN-, transformer-, and Mamba-based models, enabling a comprehensive evaluation of the proposed approach across different network paradigms. The quantitative results on different datasets are summarized in Tables II–V, while the corresponding qualitative comparisons are presented in Figs. 6–9.

1) Performance Comparison on the Vaihingen Dataset: As presented in Table II, the proposed DeepKANSeg achieved

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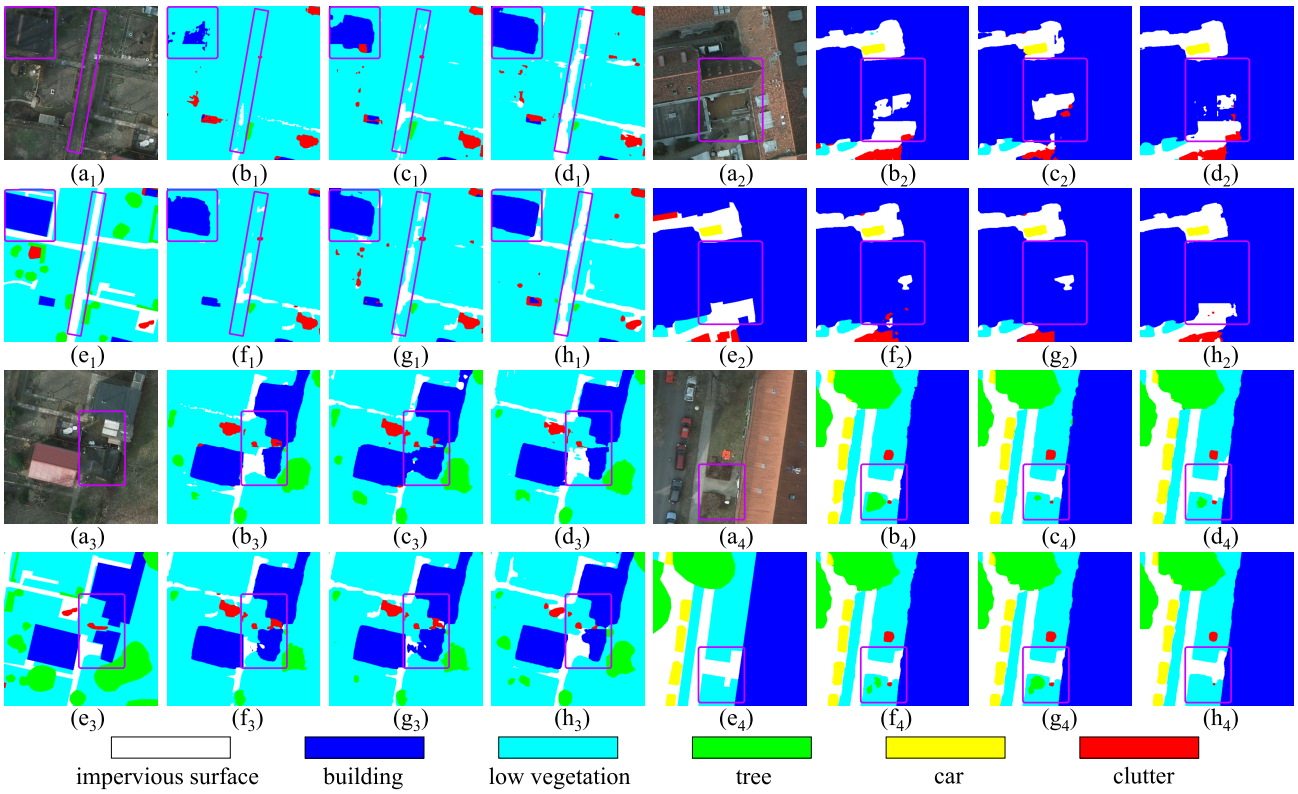


Fig. 7. Visualization results on the ISPRS Potsdam dataset of size 512 × 512. (a) RGB images. (b) TransUNet. (c) UNetformer. (d) CMTFNet. (e) Ground truth. (f) SSNet. (g) RS^{3}Mamba. (h) Proposed DeepKANSeg. Some purple boxes are marked to highlight the differences.

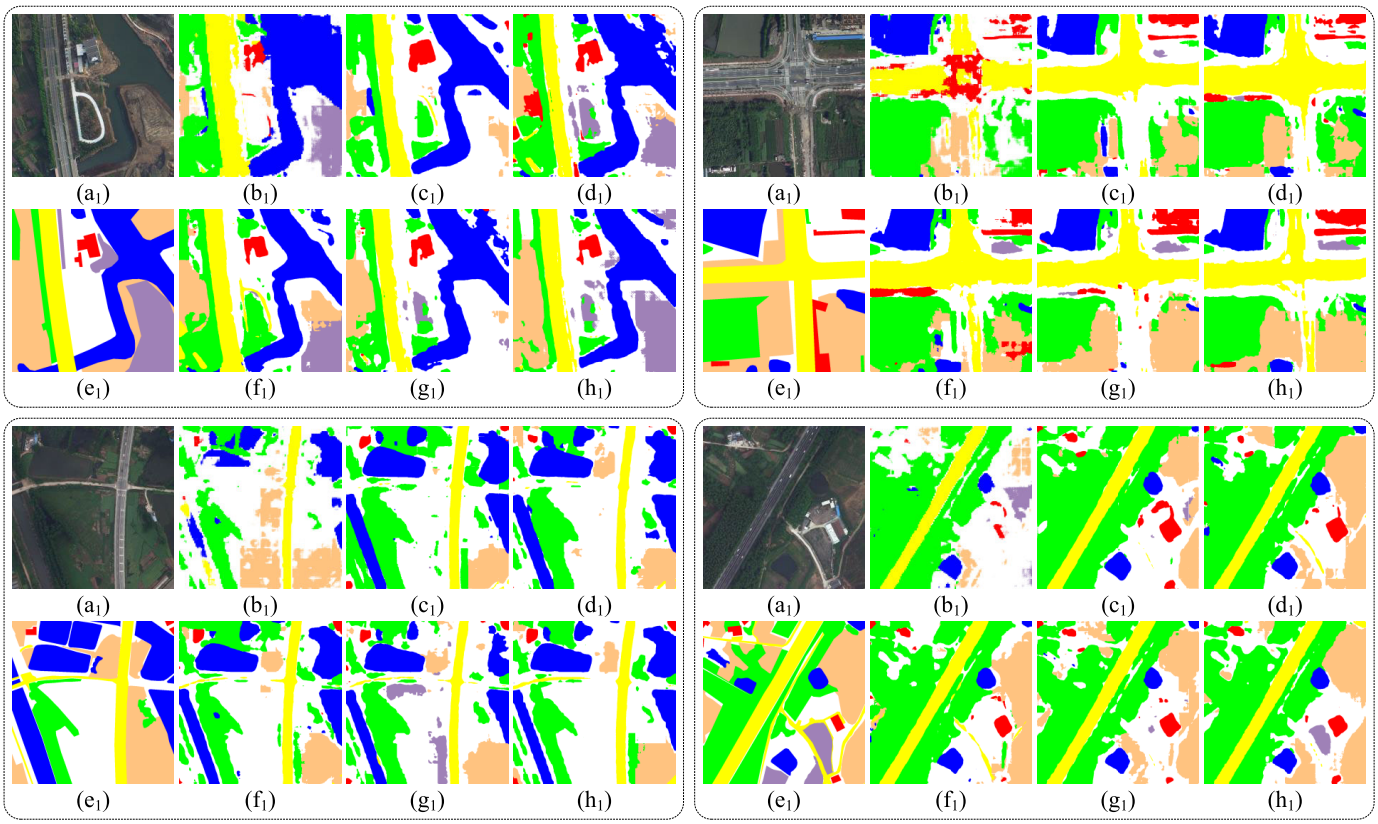


Fig. 8. Visualization results on the LoveDA dataset of size 256 × 256. (a) RGB images. (b) ACBNet. (c) CMTFNet. (d) TransUnet. (e) Ground truth. (f) RS^{3}Mamba. (g) UNetformer. (h) Proposed DeepKANSeg.

significant performance improvements in both F_1 and IoU metrics as compared to those existing methods. Specifically, DeepKANSeg, endowed with ResNet-18, attained notable improvements of 0.56% in mF_1 and 0.95% in mIoU over the

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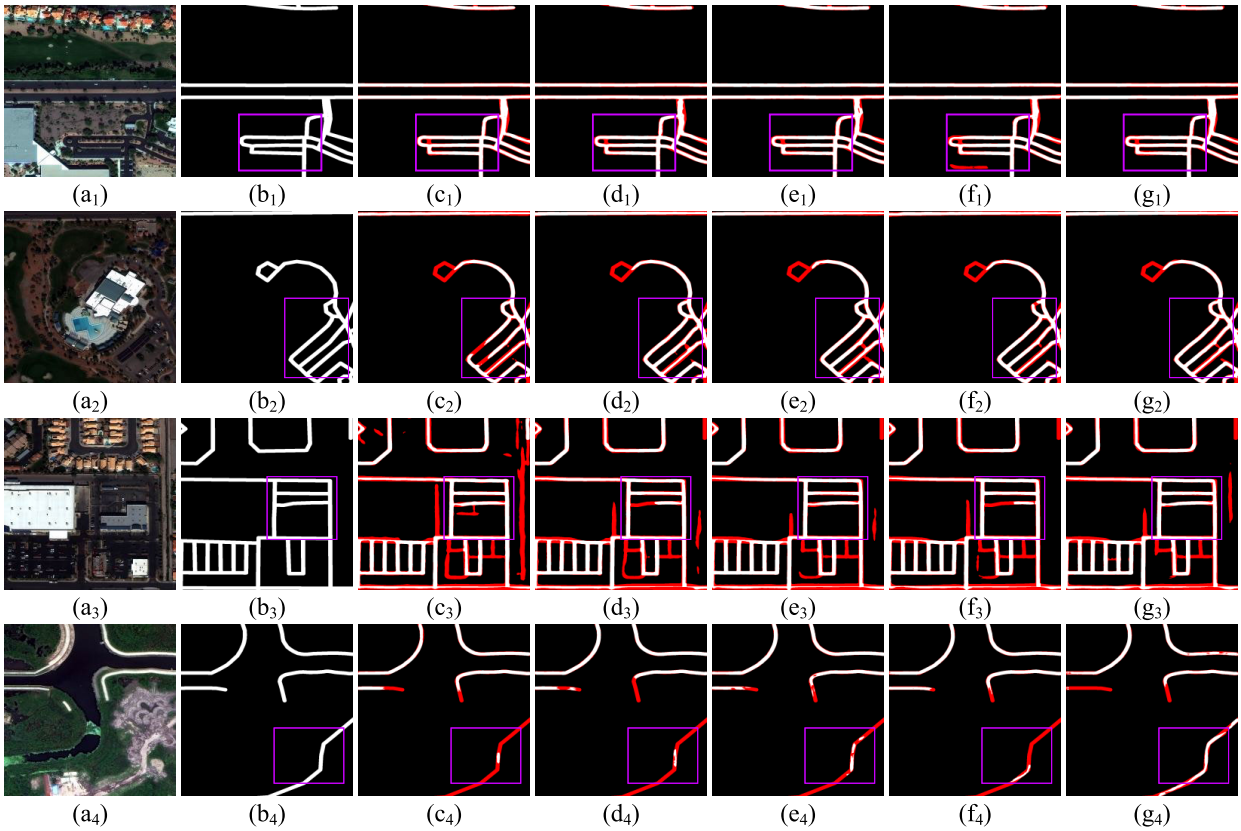


Fig. 9. Visualization results on the SpaceNet3 dataset of size 1024 × 1024. (a) RGB images. (b) Ground truth. (c) TransUNet. (d) UNetformer. (e) CMTFNet. (f) RS^{3}Mamba. (g) Proposed DeepKANSeg. Red pixels indicate incorrectly predicted regions, and purple boxes are marked to highlight the differences.

TABLE II

QUANTITATIVE COMPARISON RESULTS ON THE ISPRS VAIHINGEN DATASET WITH STATE-OF-THE-ART METHODS. HERE, WE PRESENT THE ACCURACY OF EACH CATEGORY IN F_1 AND IOU FORMAT. BOLD VALUES DENOTE THE BEST PERFORMANCE

Method	Backbone	impervious surface	building	low vegetation	tree	car	mF1	mIoU
ABCNet [22]	ResNet-18	89.78/81.45	94.30/89.21	78.49/64.59	90.08/81.95	74.05/58.80	85.34	75.20
TransUNet [77]	R50-ViT-B	90.77/83.10	94.32/89.25	79.02/65.32	90.53/82.70	82.66/70.45	87.46	78.16
UNetformer [23]	ResNet-18	92.33/85.76	96.25/92.78	80.47/67.33	90.85/83.22	89.35/80.75	89.85	81.97
FTUNetformer [23]	Swin-Base	93.41/87.64	96.92/94.02	81.53/68.82	90.91/83.33	88.46/79.31	90.24	82.62
CMTFNet [24]	ResNet-50	92.53/86.09	96.95/94.09	79.98/66.64	90.22/82.19	89.87/81.60	89.91	82.12
SSNet [46]	ViT-B	92.06/86.08	96.79/93.79	80.53/67.40	91.07/83.60	88.10/78.73	89.87	82.04
RS^{3}Mamba [30]	R18-MambaT	92.83/86.62	96.82/93.83	80.84/67.84	91.10/83.66	90.09/81.97	90.34	82.78
DeepKANSeg	ResNet-18	92.90/86.74	96.89/93.97	80.58/67.48	90.99/83.47	90.68/82.94	90.41	82.92
	ViT-L	**93.46/87.72**	**97.03/94.24**	**82.93/70.83**	**92.09/85.34**	**90.18/82.11**	**91.14**	**84.05**

baseline UNetformer. In particular, DeepKANSeg delivered substantial gains in individual classes, with improvements of 0.98% on the *Imp.* class, 1.19% on the *Bui.* class, and 2.19% on the *Car* class in terms of IoU. These enhancements indicate that the proposed DeepKANSeg effectively exploits and leverages detailed information for more accurate category predictions. Furthermore, DeepKANSeg, endowed with ViT-L, outperformed existing methods across all categories. For instance, the performance on F_1 and IoU for the *Low.* class was improved by 1.4% and 2.01%, respectively, as compared to FTUNetformer. Similarly, the performance on F_1 and IoU for the *Tre.* class was enhanced by 0.99% and 1.68%, respectively, as compared to RS^{3}Mamba.

These improvements highlight the unique strengths of DeepKANSeg in various scenarios. First, with a unified backbone, the KAN-based module effectively captured finer details of ground objects, particularly excelling in segmenting well-defined and regular structures such as the *Imp.*, *Bui.*, and *Car* classes. Second, with a more advanced backbone like ViT-L, DeepKANSeg demonstrated superior performance across all metrics, generating particularly significant gains in challenging categories such as *Low.* and *Tre.*. While existing methods performed well on segmenting simpler categories, DeepKANSeg excelled in extracting nuanced details, providing a robust and reliable foundation for accurately segmenting complex categories. Overall, the proposed DeepKANSeg endowed with

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TABLE III

QUANTITATIVE COMPARISON RESULTS ON THE ISPRS POTSDAM DATASET WITH STATE-OF-THE-ART METHODS. HERE, WE PRESENT THE ACCURACY OF EACH CATEGORY IN F_1 AND IOU FORMAT. BOLD VALUES DENOTE THE BEST PERFORMANCE

Method	Backbone	impervious surface	building	low vegetation	tree	car	mF1	mIoU
ABCNet [22]	ResNet-18	90.05/81.90	94.53/89.62	81.87/69.31	79.61/66.12	92.30/85.70	87.67	78.51
TransUNet [77]	R50-ViT-B	92.06/85.28	95.95/92.22	85.05/73.99	85.28/74.33	95.13/90.71	90.69	83.31
UNetformer [23]	ResNet-18	92.40/85.87	96.54/93.32	85.34/74.43	86.07/75.55	95.59/91.56	91.19	84.15
FTUNetformer [23]	Swin-Base	88.64/79.60	93.25/87.36	80.74/67.70	76.81/62.35	92.29/85.68	86.35	76.54
CMTFNet [24]	ResNet-50	92.46/85.97	96.81/93.82	86.30/75.90	**86.95/76.91**	96.11/92.51	91.73	85.02
SSNet [46]	ViT-B	91.98/85.16	96.58/93.38	85.32/74.40	85.93/75.33	95.78/91.91	91.12	84.02
RS^{3}Mamba [30]	R18-MambaT	92.37/85.83	96.87/93.94	85.42/74.55	86.23/75.79	95.90/92.12	91.36	84.45
DeepKANSeg	ResNet-18	92.40/85.87	97.06/94.29	86.24/75.80	85.75/75.05	95.90/92.13	91.47	84.63
	ViT-L	**92.69/86.37**	**97.11/94.39**	**86.84/76.74**	86.89/76.83	**96.30/92.87**	**91.97**	**85.44**

ViT-L achieved an impressive mF_1 of 91.14% and mIoU of 84.05%, representing increases of 0.8% and 1.27% compared to RS^{3}Mamba. These improvements can be attributed to the innovative integration of the KAN-based DFR module and the GLKAN decoder, which enable more effective extraction and refinement of high-dimensional and multiscale semantic features. In sharp contrast to the traditional methods that may struggle with complex spatial patterns in remote sensing scenes, DeepKANSeg excels in capturing detailed context, thereby addressing the challenges posed by high interclass similarity and fine-grained intraclass variations. Furthermore, the consistent enhancement across all categories highlights the robustness and generalization ability of the proposed method. These results have not only validated the effectiveness of DeepKANSeg but also demonstrated its potential for advancing semantic segmentation in complex and diverse ground environments.

Fig. 6 presents a visual comparison of the results obtained by the five best methods under consideration. Remote sensing images exhibit greater complexity than natural images, with buildings varying significantly in scale, characterized by neat borders but diverse shapes. Trees and low vegetation are often intertwined, making the segmentation task more challenging. The proposed DeepKANSeg demonstrated superior performance by capturing intricate edges with smoother results, producing more complete and connected objects. Specifically, the DFR module excels in accurately recognizing complex, long-range semantic relationships, aiding in the identification of complete objects. Furthermore, the GLKAN modules effectively extract and integrate global–local information, providing a refined foundation for final predictions. These advantages enable DeepKANSeg to deliver more precise classifications compared to other methods. Highlighted in the purple boxes across all subfigures of Fig. 6, DeepKANSeg effectively identified buildings, trees, and low vegetation amidst surrounding ground objects, resulting in cleaner and more comprehensive segmentations. This highlights its robustness in addressing the complexities of remote sensing images.

2) Performance Comparison on the Potsdam Dataset: Experiments on the ISPRS Potsdam dataset yielded results consistent with those observed on the ISPRS Vaihingen dataset, despite differences in geographical location, spectral channels, and sampling resolutions. As shown in Table III, DeepKANSeg endowed with ResNet-18 achieved mF_1 and mIoU of 91.47% and 84.63%, respectively, representing improvements of 0.28% and 0.48% over the baseline UNetformer. Significant gains were observed in three classes, with IoU improvements of 0.97% for the *Bui.* class, 1.37% for the *Low.* class, and 0.57% for the *Car* class. For the proposed DeepKANSeg endowed with ViT-L, the mF_1 and mIoU were 91.97% and 85.02%, which corresponds to increases of 0.24% and 0.42%, respectively, over CMTFNet. Notably, substantial improvements were observed in the segmentation of *Imp.*, *Bui.*, *Low.*, and *Car* classes compared to other state-of-the-art methods. Fig. 7 illustrates visualization examples from the ISPRS Potsdam dataset, with areas of interest marked by purple boxes in all subfigures. Similar to the results on Vaihingen, DeepKANSeg's predictions exhibited reduced noise points and more reliable segmentation, especially in challenging scenes.

3) Performance Comparison on the LoveDA Dataset: To further evaluate the generalization capability of the proposed method in complex remote sensing scenarios, we conducted experiments on the LoveDA dataset, which contains diverse land-cover categories and challenging spatial distributions. As shown in Table IV, DeepKANSeg achieved competitive and stable improvements over existing methods across most categories. Specifically, DeepKANSeg, endowed with ResNet-18, achieved an mF_1 of 60.34% and mIoU of 44.27%, outperforming several representative methods such as UNetformer and RS^{3}Mamba.

When equipped with a stronger backbone, DeepKANSeg endowed with ViT-L further achieved the best overall performance, with mF_1 and mIoU reaching 63.97% and 48.64%, respectively. Notably, significant improvements were observed in challenging categories such as *building*, *water*, *forest*, and *agricultural*, where DeepKANSeg consistently outperformed other methods in both F_1 and IoU metrics. These categories typically involve complex spatial patterns and high intraclass variability, demonstrating the effectiveness of the proposed KAN-based modules in modeling intricate feature relationships.

Fig. 8 presents qualitative comparisons on representative samples from the LoveDA dataset, with regions of interest highlighted by purple boxes. Compared with other methods, DeepKANSeg produces more coherent segmentation results with clearer object boundaries and reduced noise. In particular, it shows superior performance in distinguishing adjacent

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TABLE IV

QUANTITATIVE COMPARISON RESULTS ON THE LOVEDA DATASET WITH STATE-OF-THE-ART METHODS. HERE, WE PRESENT THE ACCURACY OF EACH CATEGORY IN F_1 AND IOU FORMAT. BOLD VALUES DENOTE THE BEST PERFORMANCE

Method	Backbone	background	building	road	water	barren	forest	agricultural	mF1	mIoU
ABCNet [22]	ResNet-18	48.72/32.20	59.17/42.02	52.13/35.26	58.06/40.91	36.98/22.69	51.88/35.03	30.39/17.92	48.19	32.29
TransUNet [77]	R50-ViT-B	51.27/34.47	70.37/54.28	70.19/54.07	79.81/66.41	27.93/16.23	56.79/39.65	42.01/26.59	56.91	41.67
UNetformer [23]	ResNet-18	55.24/38.16	72.39/56.73	69.73/53.53	78.26/64.28	37.73/23.25	55.78/38.68	48.61/32.10	59.68	43.82
FTUNetformer [23]	Swin-Base	55.34/38.25	75.50/60.65	73.24/57.78	72.07/56.34	38.99/24.22	56.52/39.39	51.80/34.96	60.50	44.51
CMTFNet [24]	ResNet-50	55.23/38.15	74.86/59.81	**73.90/58.60**	79.52/66.00	44.68/28.77	55.46/38.37	50.25/33.55	61.98	46.18
SSNet [46]	ViT-B	54.70/37.66	73.05/57.54	72.45/56.79	76.90/62.47	35.60/21.65	58.55/41.40	48.00/31.58	59.89	44.16
RS^{3}Mamba [30]	R18-MambaT	54.84/37.78	72.15/56.43	72.14/56.42	76.22/61.58	29.41/17.24	60.56/43.44	55.03/37.96	60.05	44.41
DeepKANSeg	ResNet-18	54.32/37.29	73.44/58.03	73.29/57.84	73.99/58.72	**45.19/29.19**	55.04/37.97	47.12/30.82	60.34	44.27
	ViT-L	**58.79/41.64**	**77.07/62.69**	73.22/57.76	**80.98/68.03**	33.07/19.81	**61.27/44.16**	**63.37/46.38**	**63.97**	**48.64**

TABLE V

QUANTITATIVE COMPARISON RESULTS ON THE SPACENET3 DATASET WITH STATE-OF-THE-ART METHODS. WE REPORT PRECISION, RECALL, F_1, AND IOU METRICS. BOLD VALUES DENOTE THE BEST PERFORMANCE

Method	Backbone	Precision	Recall	F1	IoU
ABCNet [22]	ResNet-18	65.10	55.80	60.10	42.97
TransUNet [77]	R50-ViT-B	75.20	72.60	73.90	58.57
UNetformer [23]	ResNet-18	**81.00**	65.30	72.30	56.61
CMTFNet [24]	ResNet-50	79.40	71.80	75.40	60.47
SSNet [46]	ViT-B	79.40	70.60	74.74	59.67
RS^{3}Mamba [30]	R18-MambaT	80.40	71.30	75.60	60.74
DeepKANSeg	ResNet-18	79.60	72.20	75.70	60.96
	ViT-L	78.70	**73.10**	**75.80**	**61.06**

TABLE VI

ABLATION STUDY OF THE PROPOSED DEEPKANSEG WITH TWO DIFFERENT BACKBONES ON VAIHINGEN DATASET

Backbone	DeepKAN	GLKAN	mF1(%)	mIoU(%)
ResNet-18			89.85	81.97
	✓		90.27	82.68
		✓	90.18	82.51
	✓	✓	**90.41**	**82.92**
ViT-L			90.61	83.21
	✓		90.95	83.74
		✓	90.79	83.44
	✓	✓	**91.14**	**84.05**

categories and preserving fine-grained structures in complex scenes, such as irregular agricultural areas and mixed land-cover regions. These results indicate that the DFR module effectively refines high-dimensional semantic features, while the GLKAN decoder enhances the recovery of spatial details during decoding. Overall, the consistent improvements in both quantitative and qualitative evaluations demonstrate the robustness and strong generalization ability of DeepKANSeg in complex remote sensing scenarios.

4) Performance Comparison on the SpaceNet3 Dataset: To further validate the effectiveness of the proposed method in practical applications, we evaluate DeepKANSeg on the SpaceNet3 dataset for road extraction, which can be regarded as a binary semantic segmentation task with strong structural constraints. As reported in Table V, DeepKANSeg achieves competitive performance across all evaluation metrics. Specifically, DeepKANSeg, endowed with ResNet-18, attains F_1 and IoU scores of 75.70% and 60.96%, respectively, outperforming several representative methods such as CMTFNet and RS^{3}Mamba.

With the ViT-L backbone, DeepKANSeg further achieves the best overall performance, with Recall, F_1, and IoU reaching 73.10%, 75.80%, and 61.06%, respectively. Notably, the improvement in recall indicates that the proposed method is able to better capture continuous road structures and reduce missed detections, which is crucial for road extraction tasks. These results demonstrate that the KAN-based modules can effectively model long-range dependencies and structural relationships in high-resolution imagery.

Fig. 9 shows qualitative comparisons of road extraction results, where challenging regions are highlighted with purple boxes. It can be observed that DeepKANSeg produces more continuous and complete road networks compared with other methods, particularly in complex intersections and thin road segments. In contrast, existing methods often suffer from fragmentation or discontinuities in these regions. Benefiting from the refined feature modeling of the DFR module and the global–local integration of the GLKAN decoder, DeepKANSeg is able to preserve fine structural details and maintain topological consistency. These advantages make it more suitable for real-world applications requiring accurate and reliable road network extraction.

E. Ablation Study

To assess the effectiveness of each proposed component in DeepKANSeg, ablation experiments were performed with two different backbones while preserving the encoder–decoder framework. As outlined in Table VI, these experiments were designed to examine the specific contributions of the KAN-based modules.

As shown in Table VI, the baselines were established in the first rows. When using ResNet-18 as the backbone, the baseline corresponds to UNetFormer. For the second configuration, the encoder was replaced with ViT-L, and an adapter-based fine-tuning strategy was applied, enabling the encoder to efficiently extract features from remote sensing images. In the second row of each configuration, DeepKAN modules were introduced after the encoder to evaluate their individual effect, while in the third row of each configuration, the decoder was upgraded to the GLKAN-based version to evaluate its contribution. These two configurations provide insights into the distinct roles played by the DeepKAN and GLKAN modules. The final rows represent the complete DeepKANSeg networks, integrating both DeepKAN and GLKAN modules.

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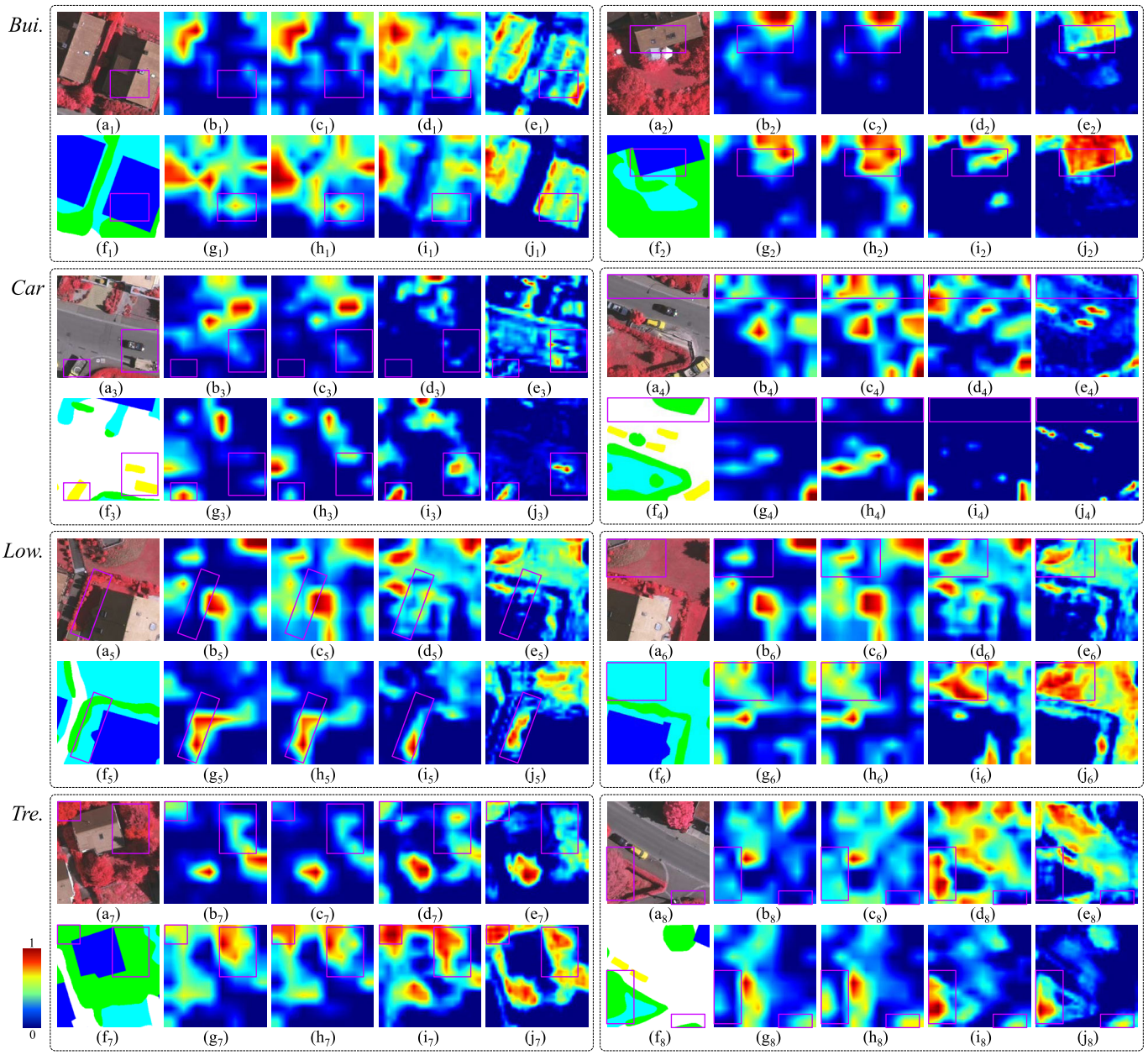


Fig. 10. Eight groups of heatmap samples are presented. In each subfigure: (a_ℓ) represents the NIRRG image, (b_ℓ)–(e_ℓ) depict four heatmaps generated by UNetformer, (f_ℓ) shows the ground truth, and (g_ℓ)–(j_ℓ) illustrate four heatmaps produced by DeepKANSeg (ResNet-18), where $\ell \in \{1, 2, 3, 4, 5, 6, 7, 8\}$ corresponds to eight distinct samples. The first, second, third, and fourth pairs of samples demonstrate how the two models determine whether a pixel belongs to *Bui.*, *Car*, *Low.*, and *Tre.*, respectively. Purple boxes have been added to emphasize the key differences.

Inspection of Table VI reveals that both the DeepKAN and GLKAN modules are essential for the improved performance of the proposed DeepKANSeg, with the contribution of DeepKAN being particularly pronounced. This highlights the critical role of DFR in remote sensing tasks. Furthermore, by effectively extracting and refining high-dimensional features, the subsequent semantic recovery process becomes more streamlined and accurate. The ablation experiments also validated the rationale behind the framework designation. DeepKAN leverages its KAN-based fine-grained learning and stacked structure to refine abstract high-dimensional semantic features, proving highly effective in capturing the intricate details and complexities characteristic of remote sensing images. Meanwhile, the GLKAN-based decoder enables comprehensive exploitation of global–local features, ensuring more precise predictions for objects with diverse spatial scales. The complementary roles of these modules are evident, and their combination ensures the robustness of DeepKANSeg, achieving superior performance in remote sensing image semantic segmentation tasks. Beyond segmentation accuracy, the model complexity of the proposed method and the comparative methods is reported in Table VII and discussed in Section IV-F.

To clearly illustrate the effectiveness of the DeepKAN and GLKAN modules, Fig. 10 presents heatmaps generated by the baseline UNetformer and the proposed DeepKANSeg

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TABLE VII

MODEL COMPLEXITY ANALYSIS PERFORMED WITH TWO 256 × 256 IMAGES ON A SINGLE NVIDIA GEFORCE RTX 4090 GPU. THE MIOU VALUES CORRESPOND TO THE RESULTS OBTAINED ON THE ISPRS VAIHINGEN DATASET. BOLD VALUES DENOTE THE BEST PERFORMANCE

Model	Backbone	FLOPs (G)	Params (M)	Memory (MB)	Speed (FPS)	MIoU (%)
ABCNet [22]	ResNet-18	7.81	13.67	**838**	**477.76**	75.20
TransUNet [77]	R50-ViT-B	64.55	105.32	1784	118.06	78.16
UNetformer [23]	ResNet-18	**5.87**	**11.69**	842	382.97	81.97
FTUNetformer [23]	Swin-Base	50.84	96.14	2540	85.91	82.62
CMTFNet [24]	ResNet-50	17.14	30.07	1496	207.43	82.12
SSNet [46]	ViT-B	45.22	85.13	1824	26.48	82.04
RS^{3}Mamba [30]	R18-MambaT	28.25	43.32	1824	106.54	82.78
DeepKANSeg	ResNet-18	19.08	43.50	1476	122.59	82.92
	ViT-L	16.86	56.67	3790	29.17	**84.05**

TABLE VIII

PERFORMANCE ON THE ISPRS VAIHINGEN DATASET AND MODEL COMPLEXITY ON DIFFERENT MODEL STRUCTURES WITH SAM BACKBONE. BOLD VALUES DENOTE THE BEST PERFORMANCE

Method	Backbone	mF1	mIoU	Params	FLOPs
SAM2UNet [78]	ViT-L	87.89	78.82	2.28M	2.35G
SAM-DBnet [79]	ViT-L	90.41	82.94	34.71M	29.52G
DeepKANSeg	ViT-B	90.97	82.57	16.59M	11.68G
DeepKANSeg	ViT-L	**91.14**	**84.05**	56.67M	16.86G

TABLE IX

SENSITIVITY ANALYSIS OF B-SPLINE BASIS SETTINGS ON THE ISPRS VAIHINGEN DATASET. THE SETTINGS MARKED WITH * DENOTE THE DEFAULT CONFIGURATION USED IN OUR EXPERIMENTS

G	r	mF1	mIoU	Param	FLOPs
3	3*	90.95	83.87	37.13M	8.88G
5*		**91.14**	**84.05**	43.50M	9.76G
7		91.15	84.04	49.86M	10.63G
9		90.75	83.4	56.23M	11.50G
5*	1	90.67	83.24	37.13M	8.88G
	2	90.88	83.61	40.32M	9.32G
	3*	**91.14**	**84.05**	43.50M	9.76G
	4	90.6	83.16	46.68M	10.19G

endowed with ResNet-18. For UNetformer, heatmaps were captured after the encoder and each of the three decoder blocks, while for DeepKANSeg, corresponding heatmaps were obtained from equivalent layers. The four rows of heatmaps demonstrate how the two models identify a pixel belonging to the classes *Bui.*, *Car*, *Low.*, and *Tre.*, respectively. First, it is evident that our method produced larger and more precise regions with high scores corresponding to the target classes, as observed across most samples. This demonstrates that DeepKANSeg effectively extracts category-specific features more comprehensively and thoroughly through refined learning. In addition, our model exhibited a stronger capability to discern differences between classes. For example, in the highlighted box of the fourth sample, our approach avoided focusing on objects that were irrelevant to the target class. This indicates that the model possesses a robust resistance to noise and distractions. These capabilities stem from the refinement capability of the KAN-based modules, which substantially enhances performance in remote sensing image semantic segmentation tasks.

In addition, to validate that the performance gains are primarily attributed to the proposed KAN-based design rather than the backbone itself, we conduct an ablation study using different model structures with ViT-based backbones, as shown in Table VIII. Specifically, SAM2UNet [78] and SAM-DBnet [79] are representative SAM-based segmentation architectures that adopt ViT-L as the encoder backbone. Under the same ViT-L backbone setting, DeepKANSeg achieves the best performance, with an mF_1 of 91.14% and an mIoU of 84.05%. Compared with SAM2UNet, DeepKANSeg improves mF_1 by 3.25% and mIoU by 5.23%; compared with SAM-DBnet, it improves mF_1 by 0.73% and mIoU by 1.11%. In terms of model complexity, DeepKANSeg with ViT-L requires 56.67-M parameters and 16.86G floating point operations (FLOPs). Although its parameter number is higher than those of SAM2UNet and SAM-DBnet, its FLOPs are lower than those of SAM-DBnet, which requires 29.52-G FLOPs. Considering the superior segmentation accuracy achieved under the same ViT-L backbone, these results indicate that the proposed KAN-based modules can effectively enhance feature representation and semantic refinement while maintaining reasonable computational complexity. In addition, DeepKANSeg is also evaluated using ViT-B as the backbone, achieving an mF_1 of 90.97% and an mIoU of 82.57%. This result exceeds other ViT-B-based methods reported in Table II, and also substantially surpasses SAM2UNet while approaching the performance of SAM-DBnet. These results further confirm that the superiority of DeepKANSeg mainly stems from its architectural design rather than reliance on stronger encoders.

Furthermore, to investigate the influence of the B-spline basis settings in the KAN layer, we conduct a sensitivity analysis on the ISPRS Vaihingen dataset, as shown in Table IX. As described in (2), the learnable univariate functions in KAN are parameterized by B-spline basis functions, where G denotes the grid size and r denotes the spline order. Therefore, we evaluate different settings of G and r to examine their effects on segmentation accuracy and model complexity. When the spline order is fixed as $r = 3$, increasing the grid size from $G = 3$ to $G = 5$ improves the mIoU from 83.87% to 84.05%. However, further increasing G brings no clear performance gain, while the model parameters and FLOPs continue to increase. Similarly, when the grid size

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is fixed as $G = 5$, increasing the spline order from $r = 1$ to $r = 3$ improves the mIoU from 83.24% to 84.05%, whereas setting $r = 4$ leads to a performance decrease. These results indicate that insufficient basis complexity may limit the representation capability of KAN, while overly complex basis settings may introduce additional computational cost and optimization difficulty. Overall, $G = 5$ and $r = 3$ achieve the best tradeoff between segmentation accuracy and model complexity, and are therefore adopted as the default setting in our experiments.

F. Model Complexity Analysis

The model complexity of the proposed method and comparative methods is evaluated using several key metrics: FLOPs, model parameters, memory footprint, and running speed. FLOPs measure the computational complexity of the model, while model parameters and memory footprint evaluate the model's scale and memory requirements, respectively. Running speed quantifies the time cost during both the training and inference stages. Ideally, an efficient method achieves lower FLOPs, fewer model parameters, a reduced memory footprint, and faster running speed. Compared with conventional MLP-based layers that rely on fixed activation functions, KAN introduces learnable univariate functions to perform more flexible nonlinear feature transformations, which may increase the number of functional parameters. Nevertheless, the proposed KAN-based modules still operate mainly along the channel dimension and do not involve the computationally intensive attention mechanisms used in Transformers or the sequence modeling operations adopted in Mamba-based methods. When DeepKANSeg uses ViT-L as the backbone, only the fine-tuned components are included in the complexity and parameter calculations.

Table VII presents the complexity analysis results for all methods considered in this work. Inspection of Table VII reveals that while our proposed model introduced some additional complexity compared to the baseline UNetFormer, it achieved significant performance improvements. Specifically, the proposed DeepKANSeg, endowed with ResNet-18, exhibited FLOPs and memory usage comparable to CMTFNet, and parameters and running speed comparable to RS^{3}Mamba. This indicates that incorporating KAN-based modules has an impact similar to adding more stacked modules in ResNet or introducing a Mamba-based auxiliary branch, as seen in RS^{3}Mamba. However, in terms of mIoU performance, our approach outperformed both methods. When utilizing a larger backbone, such as ViT-L, the proposed DeepKANSeg exhibited moderately higher complexity but delivered a substantial improvement in segmentation performance. This highlights that the KAN-based modules effectively leverage the richer features extracted by larger backbones. By offering DeepKANSeg with two different backbone configurations, we ensure that the proposed method provides superior scalability and segmentation performance, accommodating diverse application requirements compared to existing methods.

G. Discussions

1) Key Findings: The experimental results demonstrate that the proposed DeepKANSeg improves segmentation performance across different backbones and datasets. In particular, the results on the ISPRS Vaihingen and Potsdam datasets show that integrating KAN-based modules enhances the modeling of complex feature relationships in high-resolution remote sensing imagery. The DeepKAN module refines high-dimensional semantic features extracted by the encoder through fine-grained nonlinear transformations along the channel dimension, while the GLKAN decoder reconstructs spatial structures by combining global–local attention with KAN-based operations. The ablation study confirms that both modules contribute to the performance improvement, with DeepKAN playing an important role in refining semantic representations. The visualization results further indicate that the proposed framework can generate more coherent object regions and clearer structural details in complex remote sensing scenes.

2) Relationship With Existing Methods: Compared with existing semantic segmentation approaches, DeepKANSeg introduces a different perspective for modeling feature representations in remote sensing imagery. CNN-based methods mainly rely on convolutional operations to extract local spatial features, while transformer-based models capture long-range dependencies through attention mechanisms. More recently, Mamba-based approaches focus on efficient sequence modeling. In contrast, the proposed framework enhances nonlinear feature transformations within the encoder–decoder structure by leveraging the KAN. By replacing conventional MLP layers with learnable univariate functions, KAN provides a flexible mechanism for modeling complex relationships among high-dimensional features, offering a complementary alternative to existing segmentation architectures.

3) Limitations and Future Research Directions: Despite the promising results, several limitations remain. First, the proposed framework still relies on pretrained CNN or transformer encoders because efficient KAN-based backbones are not yet available. Designing dedicated KAN-based encoders may further unlock the potential of KAN in remote sensing tasks. Second, current KAN implementations often rely on B-spline functions, which may not be fully optimized for GPU parallel computation, potentially increasing computational complexity. Developing more efficient functional approximation mechanisms could improve the efficiency of KAN-based models. Finally, although KAN offers strong interpretability, analyzing learned functions in high-dimensional feature spaces remains challenging. Future work may explore more effective interpretability methods to better understand the role of KAN-based modules in deep segmentation models.

V. Conclusion

In this work, we propose a novel network, DeepKANSeg, for semantic segmentation of remote sensing images, incorporating two KAN-based modules, namely DeepKAN and GLKAN. DeepKAN leverages deep features with rich geographical information, while GLKAN

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recovers spatial relationships using global–local information. These KAN-based modules enable refined learning, leading to accurate and robust segmentation of complex remote sensing scenes. Extensive experiments on two high-resolution datasets demonstrate the effectiveness of DeepKANSeg, showing improvements in segmentation accuracy over state-of-the-art methods. Furthermore, the adaptability of DeepKANSeg across different backbones highlights its scalability for diverse applications. We also discuss future research directions to optimize KAN-based designs and improve efficiency and transparency. This work provides a foundation for integrating KAN-based technology into remote sensing tasks and advancing both methodology and practical applications.

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